Algorithmic Construction and Exploitation of Topology-Constrained Refinement Models for Semiexperimental Equilibrium Structures

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Semiexperimental equilibrium structures combine high-resolution rotational spectroscopy with computed vibrational and electronic corrections. The numerical solution of a least-squares refinement is now routine; the chemically decisive step is often the earlier one, namely the construction and validation of the inverse model that connects precise observables with structural parameters. Active coordinates, symmetry, weakly determined directions, constraints, predicates, and parameter classes are still frequently selected by expert intervention, especially for cyclic, fused, and bridged molecular graphs.

This paper introduces MORPHEUS (Model Optimization and Refinement from Perceived Hierarchy, Equivalence, Uncertainty, and Symmetry), a framework that turns model formulation into an auditable topology-perception problem. The semiexperimental refinements reported here provide the principal validation, but the construction is more general: it builds constrained refinement models from topology, symmetry, rank, isotope coverage, and external reference information before any numerical least-squares solution is attempted. Molecular topology defines the primitive coordinate space, symmetry selects the admissible active subspace, isotopic coverage determines which directions are experimentally supported, and electronic-structure reference information enters only through explicit primitive-coordinate constraints, predicates, or parameter classes.

Two equivalent active spaces are available: symmetry-adapted generalized internal coordinates built from homogeneous primitive families, and symmetry-adapted Cartesian displacements after removal of translations and rotations. The chemical interface remains internal-coordinate based, so bond, angle, dihedral, out-of-plane, ring, and local-frame relations have the same meaning whichever numerical basis is used. Validation on acyclic, cyclic, planar, fused, bridged, and flexible systems shows that rank, effective dimension, residuals, structural parameters, and propagated uncertainties are reproduced without manual coordinate-model formulation. In the norbornane-derived series, norcamphor probes a bridged reduced-dimensionality model and camphor adds methyl substitution as a stricter validation test: the accepted no-Kraitchman BDPCS3-predicate model gives sub-milliangstrom bond corrections, whereas superficially competitive variants based on unsupported hydrogen directions or tight Kraitchman predicates are rejected because they distort the geometry. The testosterone case study then shows that accurate quantum chemical equilibrium geometries, accelerated vibrational corrections, and algorithmically generated topology-constrained refinement models can be chained for molecular systems containing about 50 atoms, where isotope-rich data sets required by conventional refinement strategies are generally unavailable.

Keywords: semiexperimental equilibrium structures, model validation, generalized internal coordinates, topology-constrained refinement, Kraitchman equations

I. INTRODUCTION

Accurate equilibrium molecular structures are central to rotational spectroscopy, thermochemistry, force-field development, and the validation of electronic-structure methods.[1,2] Despite decades of progress in semiexperimental refinement, model construction remains largely a manual activity. Structure determination from experimental observables is a coordinate-dependent inverse problem: the data constrain a Cartesian geometry, but the fitted variables determine conditioning, constraints, correlations, and uncertainties. Rotational spectroscopy is a stringent example because rotational constants probe the molecular mass distribution with high precision, whereas semiexperimental (r_e^{SE}) structures require vibrational and sometimes electronic corrections to convert ground-state constants into equilibrium information.[3–6]

Modern fitting programs and workflows[7–9] have established robust statistical principles: fit well-conditioned observables, include theoretical predicates when experimental information is incomplete, and propagate covariance to derived structural parameters.[10] Their common prerequisite is that the refinement model has already been formulated. This prerequisite is especially restrictive for rings, fused systems, and bridged frameworks, where equivalent paths through the molecular graph can make one internal coordinate primary and another derived. The target of the present work is therefore not a new least-squares observable or a molecule-specific improvement in residuals, but an algorithmic route from perceived topology to a reproducible statistical refinement model.

Recent workflows have extended semiexperimental refinement by combining computed starting geometries, quantum chemical (QC) vibrational corrections, and reduced-dimensionality models.[11] The implementation

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developed here uses such data as upstream inputs while making the refinement model itself graph-defined rather than hand specified. Reference geometries from QC computations or from fragment libraries (e.g., LCB25[12]) enter only as chemically explicit constraints, predicates, or parameter classes; the fitted variables remain the algorithmically generated non-redundant active coordinates. The framework therefore converts accurate QC structures and vibrational corrections into an explicit, rank-audited refinement model without molecule-specific coordinate design.

The procedure starts from Cartesian coordinates. Primitive general internal coordinates (GICs) are generated from perceived topology, symmetry is assigned, and the totally symmetric active space is defined. Isotope coverage, symmetry orbits, chemical descriptors, and reference data then generate compatible constraints, predicates, and parameter classes. Two active spaces are available: symmetry-adapted GICs and symmetry-adapted Cartesians after removal of translations and rotations. Constraints remain primitive internal-coordinate relations and are projected onto either active basis by the Wilson matrix or the corresponding derivative operator.[13]

The rotational semiexperimental problem is used here as a demanding test because it exposes both rank deficiencies and chemically meaningful constraints. The validation set is organized so that each system probes a different failure mode of manual model formulation: acyclic references, ring closures, planar-component selection, fused and bridged topology, homologous series with different isotopic coverage, reduced-dimensionality constraints, and finally a large-molecule scale demonstration.

II. THEORY AND METHOD

A. Overview of the Framework

The overall architecture is summarized in Figure 1. The key separation is between the chemical model, the numerical active basis, and the least-squares solution. Chemical information is always expressed in primitive internal coordinates; the active variables may be GIC amplitudes or symmetry-adapted Cartesian displacements. This separation is what makes the generated model auditable: the same primitive relation has the same chemical meaning regardless of the basis used to solve the refinement.

B. Primitive and Work Coordinates

The coordinate construction starts from Cartesian geometry and perceived topology. Bonded pairs and ring connectivity define a primitive set of internal coordinates: stretches, valence angles, torsions, out-of-plane deformations, linear bends, and ring-specific primitives. Redundancy is intentional at this stage because it lets topology, not a manually chosen reference path, decide which combinations survive. For cyclic systems, endocyclic valence-angle combinations describe ring bending, whereas the endocyclic torsional manifold is reported through ring-puckering descriptors such as q and ϕ. These descriptors are constructed from the underlying dihedral combinations but are more stable and interpretable than individual endocyclic torsions, so monocyclic, fused-ring, bridged, and polycyclic systems are handled by the same machinery.[14] This primitive layer is also the chemical interface for constraints, predicates, and parameter classes.

C. Coordinates for Parameter Refinement

The active space must be non-redundant, symmetry-consistent, and compatible with chemically meaningful constraints. In the GIC route, topological primitives are reduced within homogeneous families, so stretches, bends, torsions, out-of-plane motions, and ring coordinates cannot replace one another during rank reduction.[15–19] The graph algorithms that generate the local primitive families, remove within-family redundancies, and form their symmetry-adapted combinations will be described in a separate methodological paper; here only their operational role in constrained structure refinement is needed. In particular, the present work examines how the generated coordinate spaces are coupled to isotopic coverage, rank, constraints, predicates, parameter classes, and validation diagnostics. In the Cartesian route, translations and rotations are removed from the Cartesian displacement space. In both cases symmetry selects the totally symmetric active block and completes primitive constraints over equivalent orbits.

At a reference geometry $\mathbf{x}_0$, primitive (Δs) and Cartesian (Δx) displacements are related to first order by the Wilson matrix[13]

$$\Delta\mathbf{s} = \mathbf{B}_s \Delta\mathbf{x}, \qquad \mathbf{B}_s = \left.\frac{\partial \mathbf{s}}{\partial \mathbf{x}}\right|_{\mathbf{x}_0}. \tag{1}$$

For each homogeneous primitive family α, the Wilson block $\mathbf{B}_{s_\alpha}$ is submitted to a singular-value decomposition (SVD),[20]

$$\mathbf{B}_{s_\alpha} = \frac{\partial \mathbf{s}_\alpha}{\partial \mathbf{x}}. \tag{2}$$

$$\mathbf{B}_{s_\alpha} = \mathbf{U}_\alpha \boldsymbol{\Sigma}_\alpha \mathbf{V}_\alpha^T, \tag{3}$$

The retained left-singular vectors define the row-orthonormal transformation

$$\mathbf{R}_\alpha = \mathbf{U}_{\alpha,r}^T, \tag{4}$$

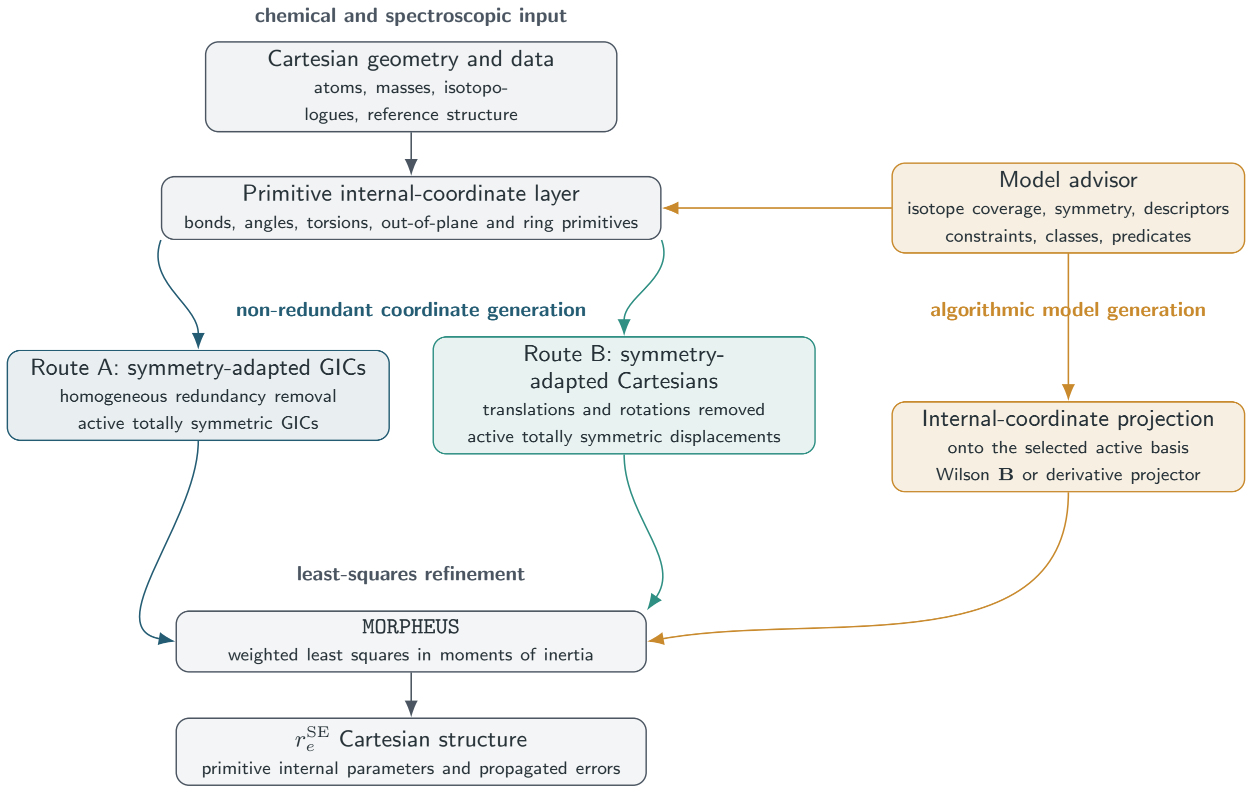


FIG. 1. Architecture of the model-generation framework.

where the subscript r denotes the retained singular subspace. The non-redundant coordinates for family α are then

$$\Delta\mathbf{q}_\alpha = \mathbf{R}_\alpha \Delta\mathbf{s}_\alpha, \qquad \mathbf{B}_{q_\alpha} = \mathbf{R}_\alpha \mathbf{B}_{s_\alpha}. \tag{5}$$

The retained rank in each family gives the number of coordinates in that family. Stacking the blocks gives

$$\Delta\mathbf{q} = \mathbf{R}\Delta\mathbf{s}, \qquad \mathbf{B}_q = \mathbf{R}\mathbf{B}_s. \tag{6}$$

The block-diagonal structure preserves chemical homogeneity. A stretch can therefore not remove a torsion from the model, and a ring coordinate cannot be lost because a different primitive family happened to enter the rank test first. After family-wise reduction, residual dependencies are pruned with deterministic topological tie breaking and the final basis is checked against the expected vibrational rank. This is a restriction on the admissible coordinate transformations, not on the rank test itself. A global SVD of the full redundant primitive set would also produce an orthonormal basis for the row space, but its left singular vectors are free to mix primitives with different chemical meaning whenever their Cartesian derivatives are linearly coupled or nearly degenerate. The numerical rank would be correct, whereas the resulting coordinates could contain arbitrary stretch–bend–torsion admixtures whose composition changes with scaling, ordering, or small geometry perturbations. Performing the rank reduction in homogeneous primitive families instead preserves the direct map between each retained coordinate and the chemical relation used for constraints, predicates, classes, and uncertainty propagation; only residual linear dependencies between already interpretable coordinates are removed in the final pruning step. Cartesian displacements associated with GIC corrections use a rank-controlled minimum-norm pseudoinverse with explicit singular-value thresholds. The symmetry-Cartesian route uses the same selection, constraint projection, and covariance propagation.

D. Constraints, Predicates, and Parameter Classes

Structural relations are specified in primitive internal coordinates,

$$\mathbf{s} = (s_1, s_2, \ldots, s_m)^T, \tag{7}$$

and may be supplied by the user or generated by the advisor. The same projector is used for constraints, predicates, classes, and error propagation, so changing from GICs to symmetry-adapted Cartesians changes the numerical basis, not the primitive chemical relations that define the model. Primitive-coordinate predicates are parsed canonically, and reversible angle or torsion orderings are normalized before projection.

a. Hard constraints. Hard constraints are exact equations imposed throughout the refinement:

$$s_i = s_i^0, \tag{8}$$

equalities,

$$s_i - s_j = 0, \tag{9}$$

or general linearized functions,

$$\mathbf{c}(\mathbf{s}) = \mathbf{d}. \tag{10}$$

At a trial geometry the corresponding first-order condition is

$$\mathbf{C}_s \Delta\mathbf{s} = \mathbf{b}_c, \qquad \mathbf{C}_s = \left.\frac{\partial \mathbf{c}}{\partial \mathbf{s}}\right|_{\mathbf{s}_0}, \qquad \mathbf{b}_c = \mathbf{d} - \mathbf{c}(\mathbf{s}_0). \tag{11}$$

Projection to the active space gives

$$\mathbf{C}_A \Delta\mathbf{a} = \mathbf{b}_c, \qquad \mathbf{C}_A = \mathbf{C}_s \mathbf{B}_s \mathbf{T}_A. \tag{12}$$

The solver keeps all compatible motions by writing the allowed correction as

$$\Delta\mathbf{a} = \Delta\mathbf{a}_0 + \mathbf{N}\Delta\mathbf{p}, \qquad \mathbf{C}_A \mathbf{N} = \mathbf{0}, \tag{13}$$

where $\Delta\mathbf{a}_0$ is a minimum-norm particular solution and $\mathbf{N}$ spans the null space. The least-squares problem is solved only for $\Delta\mathbf{p}$.

Symmetry-equivalent primitives are constrained as complete orbits before the null-space projector is formed. Reduced-dimensionality models use the same projection, allowing unsupported hydrogen frames to be frozen or tied without altering the covalent topology used for GIC generation.

b. Predicate observations. Predicates are soft structural information: they add weighted residuals rather than removing degrees of freedom.[10] A primitive predicate has the form

$$f(\mathbf{s}) = f^{\mathrm{ref}} \pm \sigma_f, \tag{14}$$

and contributes

$$r_f = \frac{f(\mathbf{s}) - f^{\mathrm{ref}}}{\sigma_f} \tag{15}$$

to the residual vector. Its Jacobian with respect to the independent least-squares variables is

$$\mathbf{J}_f = \frac{1}{\sigma_f}\frac{\partial f}{\partial \mathbf{s}} \mathbf{B}_s \mathbf{T}_A \mathbf{N}. \tag{16}$$

The numerical meaning of a predicate is therefore different from that of a constraint. A constraint removes a direction from the fit, whereas a predicate keeps the direction in the least-squares problem and assigns a stated uncertainty to the reference value. Predicate widths must consequently encode the reliability and information content of the reference geometry, not merely the desired smoothness of the refinement. Default predicate widths are intentionally conservative and reflect the expected uncertainty of the reference information rather than the intrinsic accuracy of the underlying QC method. In the automatic predicate selector used below, distance predicates are not treated as one chemically uniform class. XY bonds, where both atoms are non-hydrogen atoms, and non-CH X–H bonds are tested against the same tight propagated-error criterion because they may be directly informed by isotopic substitution, Kraitchman coordinates, or reliable local chemistry. C–H distances are kept as a separate class with a broader acceptance threshold, reflecting the frequent absence of deuterated isotopologues and the weak sensitivity of heavy-atom substitutions to local C–H directions. This distinction is a general information-content rule, not a molecule-specific exception: it prevents a model from being rejected solely because unsupported C–H coordinates have larger propagated uncertainties, while preserving a stricter criterion for the heavy-atom framework and experimentally accessible X–H bonds.

c. Parameter classes. Parameter classes make several primitive quantities share one correction. If class c contains primitives $i \in c$, the allowed correction is

$$\Delta s_i = \Delta p_c, \qquad i \in c. \tag{17}$$

Class definitions are converted into linear relations or Jacobian compression. They apply to corrections rather than final values, so related quantities may retain different reference values while sharing a fitted offset. After convergence, the covariance is expanded back to primitive and topological parameters.

In short, hard constraints remove degrees of freedom, predicates retain those degrees of freedom but add probabilistic structural information, and parameter classes reduce the parameter space by enforcing shared corrections. In the current implementation, a sensitivity advisor can then tune, but not replace, the chemical model. It evaluates the first-order response of the corrected rotational constants to each retained non-redundant GIC. Coordinates with large sensitivity, or chemically essential intermolecular character, are preferentially kept free; lower-sensitivity directions can be converted into soft predicates, and unsupported directions can be fixed. Application of these suggestions is gated by changes in rotational residuals, conditioning, and final structural displacements, and the complete decision table is written as an audit file. Thus the advisor is a conservative refinement of an already valid chemical model, not an automatic substitute for it.

E. Semiexperimental Observables

For each isotopologue s, the measured ground-state rotational constants are converted into equilibrium rotational constants by removing vibrational and, when necessary, electronic contributions,[3–6]

$$B_{\alpha,e}^{(s)} = B_{\alpha,0}^{(s)} - \Delta B_{\alpha,\mathrm{vib}}^{(s)} - \Delta B_{\alpha,\mathrm{elec}}^{(s)}, \qquad \alpha \in \{a, b, c\}. \tag{18}$$

The correction terms may come from any electronic-structure protocol and are supplied with uncertainties, masses, conventions, and provenance.[11]

Although rotational constants may be fitted directly, the refinement uses principal moments of inertia as default observables, because they give a smoother geometry dependence and avoid reciprocal amplification:

$$I_\alpha^{(s)} = \frac{K}{B_{\alpha,e}^{(s)}}, \tag{19}$$

where K is the standard rotational conversion constant.

For planar molecules only two principal components are independent. If the molecular plane is the ab plane,

$$I_c = I_a + I_b, \tag{20}$$

Fitting all three components would duplicate one constraint. The workflow therefore uses one pair, AB, AC, or BC, and reports the omitted component as a diagnostic. If the pair is not specified, all admissible pairs are tested; the selected pair maximizes rank and minimizes the propagated instability of the omitted moment.

The selected observables and uncertainties for all isotopologues form the experimental part of the least-squares objective.

F. Least-Squares Refinement

As usual,[1,7,21] the semiexperimental refinement is formulated as a weighted nonlinear least-squares problem. After hard constraints and parameter classes have been projected, the actual independent least-squares variables are collected in $\mathbf{p}$. Before this step, the code compares the number and rank of experimental observations with the projected active dimension. A completely free model that is underdetermined is not silently regularized into a nominal structure; it is reported as unsupported and must be stabilized by chemically explicit constraints, predicates, or parameter classes.

The residual vector is

$$\mathbf{r} = \mathbf{y}^{\mathrm{obs}} - \mathbf{y}^{\mathrm{calc}}(\mathbf{p}), \tag{21}$$

where $\mathbf{y}$ collects all experimental observables and predicate observations included in the refinement.

The weighted objective is

$$\chi^2 = \mathbf{r}^T \mathbf{W} \mathbf{r}, \tag{22}$$

where $\mathbf{W}$ contains the inverse variances of experimental and predicate observations. The assigned variances are treated as part of the statistical model, not merely as line-measurement errors. As emphasized in Demaison's analysis of spectroscopic structure determination,[21] model errors in vibrational corrections, electronic corrections, or missing structural information can dominate the experimental uncertainty of the rotational constants. MORPHEUS therefore propagates supplied correction uncertainties, allows conservative predicate variances, and can apply robust reweighting to complete isotopologue blocks rather than to individual rotational constants.

The optimization is performed using a weighted Levenberg–Marquardt procedure,

$$\left(\mathbf{J}^T \mathbf{W} \mathbf{J} + \lambda \mathbf{I}\right) \Delta\mathbf{p} = \mathbf{J}^T \mathbf{W} \mathbf{r}, \tag{23}$$

where $\mathbf{J}$ is the Jacobian matrix of the observables with respect to the independent variables $\mathbf{p}$ and λ is the damping parameter. Accepted steps are checked against the original topology, point group, irreducible-representation counts, and coordinate-family counts; topology-changing steps are rejected.

Convergence requires both parameter corrections and objective reduction to fall below thresholds. Diagnostic leave-one-isotopologue-out refits, small singular values, reduced rank, downweighted isotopologues, unstable planar pairs, and large propagated coordinate uncertainties are reported without redefining the coordinate model.

Following convergence, the covariance matrix in the independent fitted variables is obtained from

$$\mathbf{C}_p = \sigma_r^2 \left(\mathbf{J}^T \mathbf{W} \mathbf{J}\right)^{-1}, \tag{24}$$

where σ_r^2 is the reduced residual variance. These formal standard errors are interpreted together with rank, SVD, leverage, and leave-one-out diagnostics, because scarce isotopologue data and non-random model errors can make least-squares uncertainties overoptimistic.[21]

G. Error Propagation

Uncertainty propagation follows the geometry-update chain. If $\mathbf{N}$ is the null-space basis of the hard constraints and class reductions, the covariance in the active coordinate space is

$$\mathbf{C}_a = \mathbf{N} \mathbf{C}_p \mathbf{N}^T. \tag{25}$$

The corresponding Cartesian covariance is

$$\mathbf{C}_x = \mathbf{T}_A \mathbf{C}_a \mathbf{T}_A^T. \tag{26}$$

For a GIC refinement, $\mathbf{T}_A = \mathbf{B}_Q^+$; for a Cartesian refinement, $\mathbf{T}_A = \mathbf{C}_{A_1}$. Thus both coordinate choices use the same covariance propagation.

Reported structural quantities are functions of the optimized Cartesian geometry. For a vector of quantities $\mathbf{g}(\mathbf{x})$, which may contain bond lengths, valence angles, dihedral angles, out-of-plane coordinates, ring descriptors, or user-defined primitive functions, the covariance is

$$\mathbf{C}_g = \mathbf{G}_x \mathbf{C}_x \mathbf{G}_x^T, \qquad \mathbf{G}_x = \left.\frac{\partial \mathbf{g}}{\partial \mathbf{x}}\right|_{\mathbf{x}_{\mathrm{opt}}}. \tag{27}$$

Standard errors and correlations are obtained from $\mathbf{C}_g$. Hard constraints have no variance along removed directions, whereas predicate uncertainties enter the weights and contribute to $\mathbf{C}_p$.

The report lists propagated structural errors, correlations, SVD diagnostics of the final weighted Jacobian, projected constraints and classes, and matched active-coordinate labels. Final bond lengths, angles, dihedrals, and ring parameters are always evaluated from the optimized Cartesian geometry, so the reported tables are independent of whether the fit used GIC or Cartesian amplitudes. A numerical Hessian in the final independent variable space gives the stationary-point diagnostic.

H. Kraitchman Diagnostic

For singly substituted isotopologues, the workflow computes substitution coordinates from the Kraitchman equations[22] and compares their absolute values with the fitted coordinates in the parent principal-axis frame. Kraitchman coordinates are widely used in spectroscopic structure work, either directly as r_s substitution coordinates, as checks on r_0 or semiexperimental structures, or as practical atom-assignment diagnostics. In the present workflow they are treated as diagnostics by default. They may be converted into soft predicates only as explicit, weighted external structural information, and only if the resulting predicate set survives the same stability tests as any other reduced-dimensionality model. This distinction is important because Kraitchman coordinates do not retain coordinate signs and do not exploit the full correlated model. At the same experimental-information level and computational cost, the SE+fit route is therefore more informative: all substituted constants enter one constrained Cartesian fit, uncertainties are propagated through the same model, and omitted or weak components can remain external diagnostics rather than hidden constraints.

III. IMPLEMENTATION AND REPRODUCIBILITY

The implementation is organized around two reusable services. The coordinate-definition service builds the molecular coordinate model from Cartesian atoms and nuclear charges, assigning point-group symmetry and irreducible representations when requested. A separate B-matrix service evaluates the same frozen coordinate definition for any compatible Cartesian geometry. During refinement, the engine reuses the stored coordinate model and updates only the derivative information needed for the current step.

The standard input contains the Cartesian parent geometry, optional primitive freeze or constraint records, and isotopologue data with rotational constants, corrections, uncertainties, masses, and conventions.

Each run records the final Cartesian structure, propagated primitive parameters, rotational residuals, covariance and SVD diagnostics, projected constraints/classes, sensitivity-advisor decisions when requested, provenance, warnings, and minimum-check Hessian eigenvalues. These outputs provide the audit trail for the algorithmically generated model and are the numerical source for the benchmark tables.

The production report also records the final primitive parameters together with their starting values, final-minus-starting shifts, and propagated standard errors. A final-validation stage can replay the same accepted model with the alternate active coordinate basis, robust loss functions, predicate width scans, leave-predicate-group-out tests, random multistarts, leave-one-isotopologue checks, and predicate audits. These checks are not used to lower the residual. They test whether a proposed model is reproducible, well conditioned, and chemically stable under reasonable perturbations of the assumptions that made the inverse problem soluble.

IV. RELATIONSHIP TO EXISTING APPROACHES

The framework combines established semiexperimental least-squares machinery[1,7,10,11] with an algorithmic model-generation layer. The statistical model uses moments of inertia, vibrational and electronic corrections, predicates, parameter classes, leverage and correlation diagnostics, robust weighting, and final minimum checks. The new element is not a replacement for the Wilson matrix, SVD rank analysis, linearized constraint projection, or covariance propagation, which are established tools. It is their use in an automatic topology- and symmetry-driven construction of non-redundant, symmetry-adapted internal coordinates, followed by automatic generation of the reduced refinement model and its audit trail. Existing approaches automate the solution of a model; this work formalizes the generation of the model itself, and then solves it with the same statistical discipline.

Z-matrices remain compact and intuitive for tree-like molecules, but rings, fused systems, and bridged frameworks expose their reference-tree dependence.[23] GICs avoid a single reference tree, but in standard optimization contexts they are commonly redundant, delocalized, or user-defined. The present construction instead generates the GIC space from topology-based primitive coordinates, reduces it to a non-redundant space, assigns irreducible representations, and keeps constraints and predicates tied to primitive-coordinate relations. Kisiel's `STRFIT`/PROSPE route[8] remains an important reference point for rotational-structure fitting, but the structural model is again supplied by the user.[9]

The practical distinction is summarized in Table I. The workflow retains the usual semiexperimental inputs and diagnostics while moving coordinate construction, symmetry completion, constraint and class generation, and covariance propagation to a common graph, symmetry, and rank layer. The user still controls the chemical problem, the quality of the input data, and any deliberate override of the default tolerances. The production workflow, however, uses robust default thresholds for class assignment, cross-contamination rejection, predicate widths, and rank tests, so the reported benchmarks do not rely on molecule-specific coordinate editing or case-by-case active-space construction.

A. Computational Reference Data

The electronic-structure quantities used in this work were generated with the hierarchical family of Pisa composite schemes (PCSs),[12,24] employing the Gaussian suite of programs.[25].

In most cases, vibrational corrections and constraint targets were taken from the double-hybrid member of the family (DPCS3) and its bond-corrected enhancement (BDPCS3), which have been validated in previous work for accurate structures and fast rovibrational corrections.[11,26,27] The vibrational contributions to the rotational constants (ΔB_{vib}) were evaluated with the recently introduced finite-difference-of-gradients approach along normal modes, which enables efficient and accurate calculations also for medium-sized and large molecular systems.[11] For the literature benchmarks, the actual ΔB_{vib} values are taken from the cited experimental or semiexperimental data sets; the finite-difference-of-gradients protocol is used here as new large-molecule input for testosterone, with the computational recipe and convergence details documented in Ref. 11.

The only exception is conformer I of glycine, where the best available reference structure was used: PCS4, based on explicitly correlated CCSD(T)-F12 calculations close to the complete-basis-set limit with an additive core–valence correction. Outside this role as reference information, the electronic-structure data do not define the fitted coordinates; they enter only through explicit constraints, predicates, parameter classes, and ΔB_{vib} corrections.

V. RESULTS AND DISCUSSION

The main benchmark systems are shown in Figure 2 and are discussed in the order in which they test increasingly demanding parts of the same model-generation problem. The results are therefore not independent case studies but a sequence of stress tests, each designed to validate one property of the framework. Glycolaldehyde establishes the acyclic baseline; cyclopentadiene tests ring closure; nitrobenzene tests planar-component selection; azulene tests fused aromatic topology; the norbornane-derived series tests bridged graphs, first through norcamphor and then through the methyl-substituted camphor framework. The anhydride series probes how published data sets with different isotopic completeness are converted into equivalent input models. Glycine I and II test reduced-dimensionality constraints. Within the bridged series, camphor provides the focused model-validation stress test, because the additional methyl substitution increases the number of local hydrogen directions and separates a chemically admissible predicate model from alternatives that give attractive residuals but unacceptable structures. Testosterone is kept separate and last because it is a scale demonstration rather than an isotope-rich benchmark. In all cases the input is Cartesian and no molecule-specific coordinate code, manual reference tree, or manual active-space selection is used.

Figure 3 is the central validation of this logic: it quantifies how topology perception, symmetry analysis, isotopic coverage, and reference information are transformed into the final refinement model. The figure is therefore a model-generation audit, not simply a residual summary: it reports how many primitive/non-redundant coordinates are produced, how many remain active after symmetry selection, how many survive constraint projection, and how many constraints are introduced without manual parameter selection. The rotational residuals remain in the expected semiexperimental range for standard, cyclic, planar, fused, bridged, and glycine benchmarks, but the central result is that these residuals are obtained from generated, auditable models. The glycine examples also separate two roles of theory: Gly-I starts from a reference that already satisfies the practical rotational-accuracy target, whereas Gly-II uses QC-derived relations to define missing reduced-dimensionality information.

Figure 3 also makes explicit what is generated. For azulene, topology and symmetry generate 48 final GICs, identify 17 totally symmetric coordinates, and reduce the fit to 10 effective variables without a manual choice of refined coordinates. For norcamphor, the same machinery constructs a non-redundant 48-coordinate GIC model for a low-symmetry bridged system, then generates 30 hydrogen-frame constraints and leaves 18 effective variables. These are the cases in which manual model formulation is most exposed: fused rings and bridged frameworks make it difficult to decide by inspection which

TABLE I. Compact operational comparison of representative structure-refinement routes. The emphasis is model formulation rather than the statistical quality of a particular fit.

Feature	Z-matrix	User GIC	MORPHEUS
Input model	Internal tree	User model	Cartesian topology
Active variables	Manual	Manual	Rank-pruned
Rings/bridges	Tree-dependent	User-defined	Graph-derived primitives
Missing isotopes	Manual constraints	Manual classes	Generated constraints
Symmetry	In model	User supplied	Detected and audited
Output audit	Supplied model	Workflow dependent	Manifest and covariance

distances, angles, torsions, and hydrogen-frame quantities should be refined, tied, or constrained. Table II recasts the same information as an operator comparison. The comparison is not based on elapsed time, which is operator- and implementation-dependent, but on reproducible decision classes: which cyclic or bridged coordinates to use, which symmetry block to refine, which unsupported directions require constraints, which planar rotational components are admissible, and which parameters must remain active after rank projection. In a manual workflow these choices precede the least-squares solution and are a common source of non-reproducibility. In the reported runs they are made by topology perception, symmetry assignment, coverage analysis, and rank checks. Thus the comparison is a test of reproducible model formulation, not only of the numerical least-squares solution.

The agreement with reference-quality refinements is therefore obtained after removing the molecule-specific model-design step, not after tuning it.

The three anhydrides were selected as a homologous literature-data series with a common chemical motif but progressively different experimental completeness, and therefore test different parts of the model layer. Maleic anhydride is the fully determined case: the published work of Vogt, Demaison, and Rudolph[28] reports the parent geometry, ground-state rotational constants, rovibrational corrections, and semiexperimental constants for a sufficiently rich isotopic set. Phthalic anhydride tests mixed estimation: Belyakov et al.[29] supplement the rotational data with weighted r_e^{BO} predicates. Succinic anhydride tests the opposite limit: Jahn et al.[30] provide only heavy-atom substitutions, so the hydrogen frame is not experimentally resolved and must be represented by explicit model relations.

This series also makes the input claim concrete. Once B_0 constants and rovibrational corrections ΔB_{vib} are available, the user transcribes the parent Cartesian geometry, the isotope substitutions, and the tabulated corrections. The program then perceives the topology and symmetry, constructs the non-redundant coordinate model, and the model layer applies any necessary constraints, predicates, or parameter classes as explicit statistical relations.

For maleic anhydride no auxiliary relation is introduced and 21 non-redundant symmetry coordinates are generated, of which the eight A_1 coordinates are the active variables for the planar SE fit (Table III). The GIC and symmetry-Cartesian routes both give rank 8 and a rotational RMS residual of 0.036 MHz; the recovered bond lengths agree with the published r_e^{SE} values within 1.2×10^{-4} Å.

Table IV illustrates the type of chemically meaningful relations generated by the advisor. The program identifies unsupported structural directions and expresses them as primitive internal-coordinate relations, which are subsequently projected onto the algorithmically generated active space. The same representation also makes ring coordinates explicit: endocyclic bends are generated from primitive valence angles, while ring-puckering descriptors such as q and ϕ are constructed from the endocyclic dihedral manifold rather than selected as molecule-specific torsions.

For phthalic anhydride, the rotational information alone can give a small rotational residual while leaving the structural model rank deficient. The published solution is therefore naturally expressed as primitive-coordinate predicates with conservative uncertainties of 0.002 Å for bond lengths and 0.2° for angles. With these predicates, the fit has rank 14, condition number 4.05, and a rotational RMS residual of 0.50 MHz. It recovers the published r_e^{SE} structure within 0.0015 Å for the tabulated bond lengths and within 0.09° for the tabulated angles.

Succinic anhydride completes the series because only heavy atoms are isotopically substituted. The heavy-atom framework is determined, but the hydrogen positions are not independent spectroscopic observables. In this case a local hydrogen-frame constraint gives a stable fit with rank 5 and a rotational RMS residual of 0.035 MHz. Weighted C–H predicates are still useful diagnostically, but as soft observations they do not remove all unobserved hydrogen-frame directions; without an additional topological or constraint relation the optimizer can enter geometrically equivalent but chemically unacceptable valleys. The comparison makes the distinction operational: constraints remove unsupported directions, predicates add observations with declared uncertainty, and parameter classes tie equivalent corrections.

For planar molecules the model layer selects the best pair of rotational constants to be fitted, and retains the third rotational constant as an external check computed from the final Cartesian geometry. Figure 4 shows that the AB pair gives the smallest complete diagnostic resid-

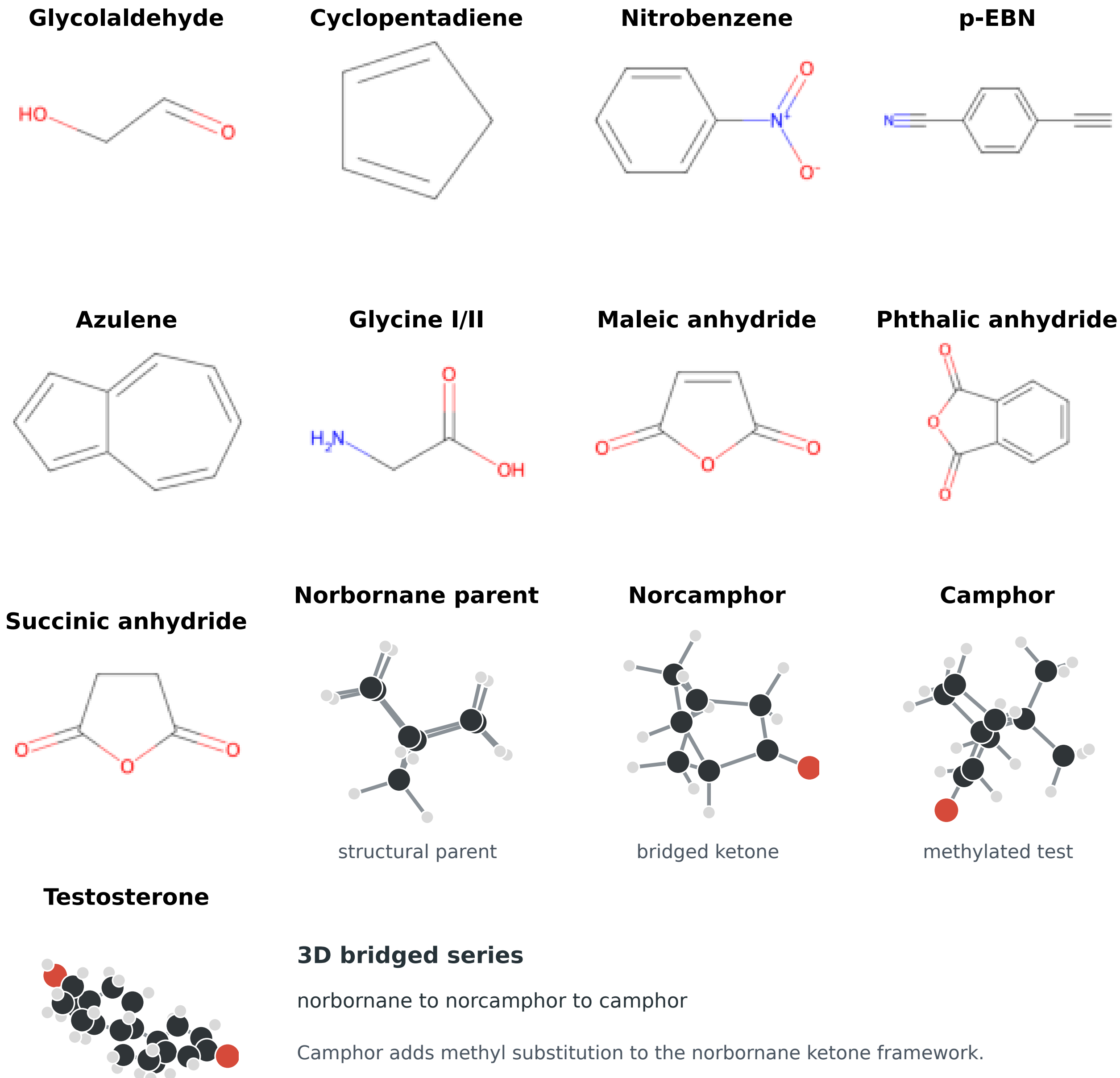


FIG. 2. Molecular structures used to test algorithmic model generation. The acyclic, planar, aromatic, glycine, and anhydride benchmarks are shown as connectivity diagrams; the bridged norbornane family and testosterone are shown as three-dimensional structures because their graph topology is not communicated adequately by a planar drawing. Norbornane is included as the structural parent of the bridged series, while norcamphor and camphor are the semiexperimental validation cases. Detailed atom numbering for the tabulated structural parameters is given in the Supporting Information.

TABLE II. Decision burden removed by algorithmic model generation. Manual setup time is not reported because it depends on operator choices; the table therefore reports the reproducible decision burden. The final column counts molecule-specific coordinate or constraint edits made by the user in the reported runs.

Case	Manual burden removed	Model	Constraints/classes	Outcome
Azulene	fused rings; symmetry; planar pair	48 GICs; 17 sym.; 10 var.	8 primitive constr.	0.073 MHz RMS; 0 edits
Norcamphor	bridged graph; H frames; covariance	48 GICs; 48 active; 18 var.	30 H-frame constr.; 58 predicates	0.094 MHz RMS; 0 edits
Glycine I	symmetry; unsupported H directions	24 GICs; 15 A'; 11 var.	4 QC constr.	0.0269 MHz RMS; 0 edits
Glycine II	low symmetry; coupled descriptors	24 GICs; 19 var.	5 QC constr.; classes	0.249 MHz RMS; 0 edits

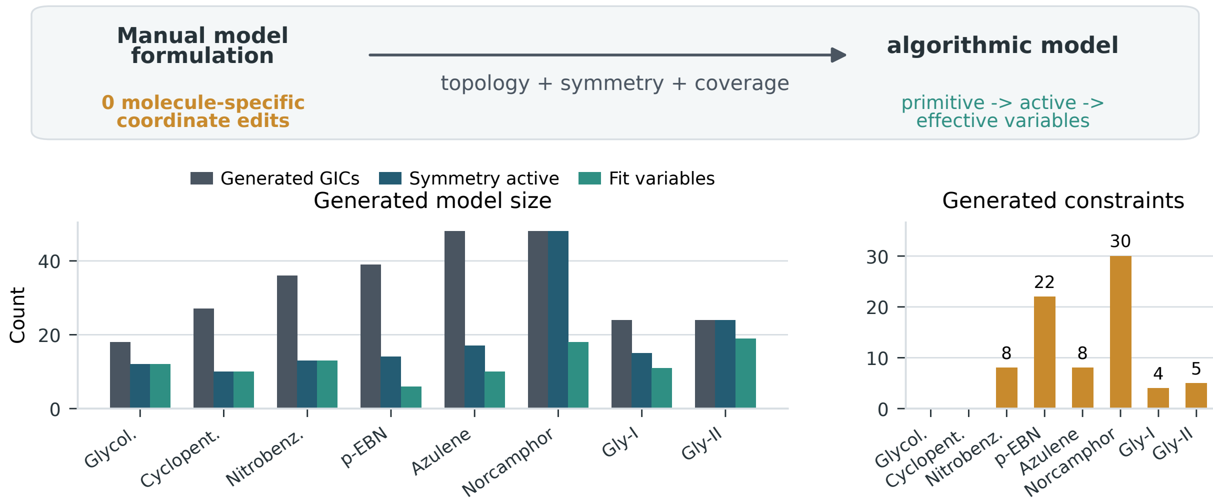


FIG. 3. Algorithmic model-generation audit for the representative refinements. The top guide links the manual formulation problem to the generated model: topology perception, symmetry analysis, and isotopic coverage replace molecule-specific coordinate edits under the default global thresholds. The left panel reports the number of coordinates generated algorithmically, the symmetry-active subset before constraint projection, and the final least-squares variables after reduction. The right panel reports the number of constraints or frozen primitive parameters projected by the model layer. Manual coordinate-model edits are zero for every case; global defaults can be inspected or overridden, but are not tuned separately for the reported molecules. Detailed fit residuals and ranks are collected in the Supporting Information.

TABLE III. Algorithmically generated symmetry-coordinate model for maleic anhydride from literature data. Only the totally symmetric A_1 block is active in the C_{2v} semiexperimental refinement.

Symmetry block	Coordinates	Role in the SE model
A_1	8	Active stretches/bends; rank 8
A_2	3	Torsion/out-of-plane diagnostics
B_1	3	Torsion/out-of-plane diagnostics
B_2	7	Stretch/bend diagnostics
Total	21	Non-redundant GIC space

TABLE IV. Chemically readable primitive-coordinate relations generated by the model layer. The first four entries are representative reduced-dimensionality constraints for glycine; the last two show the corresponding form of generated ring coordinates. Atom labels are used for readability.

Primitive relation	Generated relation	Role
O–H bond	`ROH = R(O,H)`	fixes unsupported O–H distance
C–O–H angle	`AOH = A(C,O,H)`	preserves hydroxyl orientation
H–N–H angle	`HNH = A(H,N,H’)`	constrains amino frame
NH_2 wag	`WNH2 = OOP(H,N,H’,C)`	removes unsupported wagging direction
Endocyclic bend	`RB = sum c(i) A(i,j,k)`	defines ring valence coordinate
Ring puckering	`q, phi = f(D-ring set)`	reports puckering from endocyclic dihedrals

TABLE V. Effect of auxiliary model relations for phthalic anhydride. The predicate model uses the conservative r_e^{BO} uncertainties reported by Belyakov et al. and the same literature rotational data. Structural errors are maximum absolute deviations from the tabulated r_e^{SE} bond lengths or angles.

Auxiliary relation	Rank	Cond.	RMS	Error
None	13	∞	0.0065	3.13°
Frozen H frames	9	1056	0.0067	0.19°
r_e^{BO} predicates	14	4.05	0.50	0.0015 Å; 0.09°

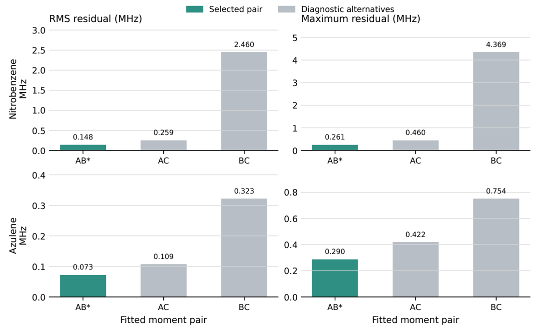


FIG. 4. Planar-component selection for nitrobenzene and azulene: the selected AB pair is highlighted and minimizes the complete post-fit diagnostic residual while the omitted component remains an external check. Detailed values are reported in the Supporting Information.

TABLE VI. Leave-one-isotopologue sensitivity diagnostics for representative refinements. The table reports exact refits in which one isotopologue is omitted, followed by prediction of the omitted rotational constants.

System	Refits	Worst omitted RMS MHz	Omitted	Max Cartesian shift Å	Cond. range
Maleic anhydride	10	0.086	13C2d2	0.0005	6.1×10^2–8.0×10^2
Nitrobenzene	9	0.388	3D	0.0840	2.6×10^4–3.7×10^5

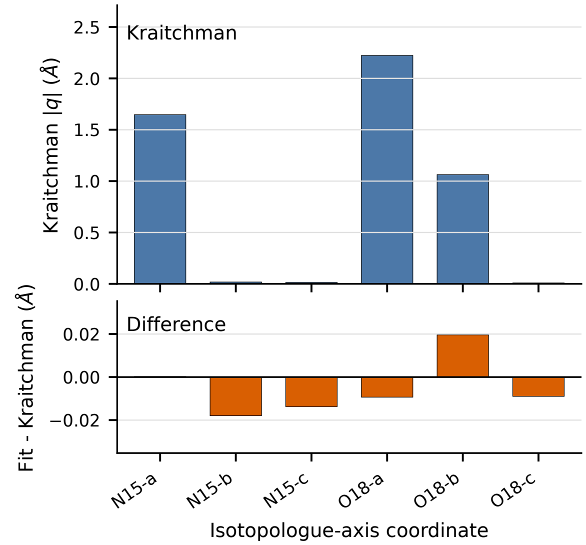


FIG. 5. Kraitchman diagnostic for the nitrobenzene planar benchmark. The upper panel reports absolute substitution coordinates from the Kraitchman analysis. The lower panel reports signed differences between the final SE coordinates and the Kraitchman values on an expanded scale. Numerical values are reported in the Supporting Information.

ual for both nitrobenzene and azulene.

The condition numbers in the benchmark table should be interpreted as diagnostics of the inverse problem, not as residual-quality measures. Nitrobenzene is the clearest example. It fits only the planar AB component pair from nine isotopologues while retaining 13 active variables and eight primitive constraints; several deuterated substitutions therefore carry substantial leverage on directions that are weakly separated by the moments of inertia. The result is a high condition number (2.75×10^4) even though the final rotational RMS residual is only 0.148 MHz. Azulene, by contrast, has a larger fused-ring coordinate space but the algorithmic reduction leaves only 10 effective variables, and the additional isotopic coverage and primitive constraints distribute the spectroscopic information more evenly. Its condition number is therefore lower (9.57×10^2) despite the more complex topology. The same reading applies to the other benchmarks: high condition numbers flag directions whose uncertainties or influence diagnostics must be inspected, whereas acceptable residuals alone do not prove that the structural model is statistically well supported.

Table VI gives a direct sensitivity check. Maleic anhydride is stable under exact leave-one-isotopologue refits: the worst omitted-isotopologue RMS is 0.086 MHz and the largest Cartesian shift is 5×10^{-4} Å. Nitrobenzene remains predictive for most omissions, but the deuterated isotopologues expose the expected weak directions: the worst omitted RMS rises to 0.388 MHz and one refit reaches a condition number of 3.7×10^5. This is precisely the regime in which the algorithmic model is useful to an operator: it does not hide the problem behind a small training residual, but reports the rank, condition number, influence, and leave-one-out diagnostics needed to decide whether additional data or stronger model relations are required.

The next two diagnostics compare the fitted structures with independent structural information without changing the least-squares objective. The nitrobenzene Kraitchman comparison in Figure 5 confirms that the fitted Cartesian structure follows the substitution-coordinate pattern, while the fit itself remains based on moments of inertia. This is the intended use of Kraitchman information here: it validates the SE+fit geometry but does not replace the statistically coupled refinement.

The Gly-I structural comparison in Figure 6 shows the scale of the semiexperimental correction when the starting QC geometry already reproduces the rotational data well. Since the reduced-dimensionality constraints are generated from the QC reference structure, the small SE–QC differences provide an a posteriori validation of the generated constraint model, not merely a check that the starting geometry was good.

Detailed atom numbering, rotational constants, topological parameters, Kraitchman diagnostics, reference-geometry comparisons, residuals, conditioning, and covariance diagnostics are reported in the Supporting Information. The main text retains the information needed to assess model formulation: rank, dimensionality, symmetry, constraint projection, planar-component choice, residual scale, and minimum checks.

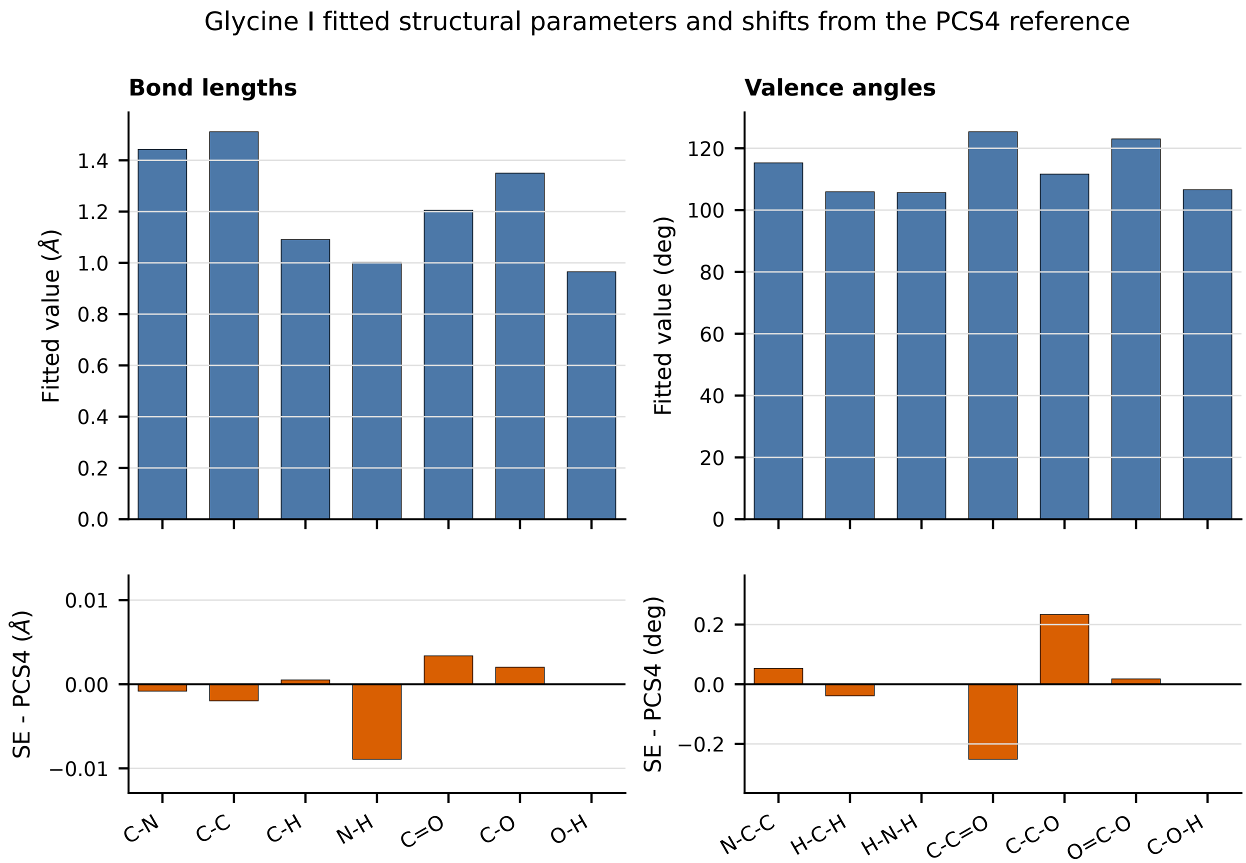


FIG. 6. Glycine I fitted structural parameters and shifts from the QC reference geometry. The upper panels report final SE bond lengths and valence angles; the lower panels report signed SE–QC differences on expanded scales.

A. Generated Reduced-Dimensionality and Validation Models

Incomplete isotopic coverage can leave some structural directions unsupported. The advisor therefore treats reduced dimensionality as part of the model-generation step. It preserves the highest compatible symmetry, identifies local coordinates that cannot be determined independently, and replaces them by primitive-coordinate constraints, predicates, or parameter classes.

Table VII shows that the reduced models are generated from isotope coverage and chemical equivalence rather than imported as a manual parameter subset. In glycine I, glycine II, and norcamphor the available substitutions do not support all hydrogen-related directions independently; the advisor therefore converts the unsupported directions into explicit primitive-coordinate relations derived from the reference structure.

The bridged cases are best viewed as a norbornane-derived progression. Norbornane defines the bicyclic topology, norcamphor adds the ketone and sparse heavy-atom substitution data, and camphor adds methyl substitution and a larger set of local hydrogen directions. Norcamphor is therefore the reduced-dimensionality member of the series: all 48 GICs belong to the totally symmetric space, so the reduction is controlled by coverage and local-frame diagnostics rather than by point-group symmetry. The updated `xyzin` input uses the experimental constants of Table 3 of the norcamphor rotational study, the tabulated vibrational corrections, and Kraitchman-derived soft predicates generated from the singly substituted heavy-atom isotopologues. The workflow uses 30 hydrogen-frame constraints and 58 Kraitchman predicates to obtain an 18-dimensional fit with a rotational RMS residual of 0.094 MHz.

Camphor then tests the complementary question: not whether unsupported directions can be stabilized, but whether a generated model can reject a tempting but chemically wrong stabilization in the same bridged family. The accepted run uses 12 isotopologue records, 36 rotational constants, a full-rank 75-variable GIC model, and 178 BDPCS3 soft predicates; no hydrogen coordinate is frozen and no Kraitchman-derived predicate is retained in the least-squares objective. The rotational RMS residual is 0.0788 MHz, with a maximum absolute residual of 0.284 MHz. More importantly, the final structural correction to the BDPCS3 reference remains small on the chemically relevant scale: the largest bond-length

TABLE VII. Autonomous structural-model decisions made by the advisor in the reduced-dimensionality and validation benchmarks. “Active variables” are the independent least-squares coordinates after symmetry selection and projection of hard primitive-coordinate constraints.

System	GIC space	Variables	Generated constraints	Classes
Glycine I	24 GICs; 15 A'	11	4 QC constraints	none
Glycine II	24 GICs; C_1	19	5 QC constraints	coupled descriptors
Norcamphor	48 GICs; C_1	18	30 H-frame; 58 Kraitchman	constrained H frames
Camphor	75 GICs; C_1	75	178 BDPCS3 predicates; no Kraitchman	validation-controlled predicates

shift is 0.832 mÅ, the largest valence-angle shift is 0.040°, and the corresponding propagated standard errors are bounded by 3.000 mÅ and 0.257°.

The validation suite makes this distinction quantitative. Replaying the same model in the symmetry-adapted Cartesian basis changes the aligned Cartesian structure by only 5.2×10^{-11} Å RMS; robust-loss and multistart checks change it by less than 5×10^{-4} Å RMS, and predicate-width scans keep the maximum atomic displacement below 9.5×10^{-4} Å. By contrast, removing hydrogen-related predicates gives lower or comparable rotational residuals but shifts atoms by up to 0.935 Å, and tight Kraitchman predicates produce bond and angle distortions up to 7.65 mÅ and 0.343°. Broad Kraitchman predicates are not damaging, but they do not add useful information to the validated BDPCS3-predicate model. The camphor member of the bridged series therefore shows the operational role of the advisor: it does not merely assemble a least-squares input, it proposes a chemically explicit model and then rejects variants that fail reproducibility, robustness, or structural-stability checks.

B. Glycine as an End-to-End Model-Generation Test

The two glycine conformers test the same model-generation layer starting from QC geometries when hydrogen-related directions are only partly supported.

For glycine I, the parent, ^{13}C(carboxyl), ^{13}C(methylene), CD_2, and ^{15}N isotopologues are fitted using the rotational constants and weights of Godfrey and Brown[31] and the zero-point corrections of Kasalová et al.[32] The QC reference already reproduces the 15 vibrationally corrected constants with an RMS relative error of 0.0258% and a maximum error of 0.0363%. The advisor detects the C_s geometry and incomplete hydrogen information and generates four constraints: O–H distance, C–O–H angle, H–N–H angle, and NH_2 wag. Projection leaves 11 independent variables from 15 A' GICs, and the final fit gives a complete rotational RMS residual of 0.0269 MHz with a positive optimized-space Hessian.

The same Gly-I model was also rerun with weighted initial-geometry predicates on XY and X–H distances and selected heavy-atom/C–O–H angles ($\sigma_R = 0.003$ Å, $\sigma_A = 0.3°$). The rank remains 11, the condition number

TABLE VIII. Glycine I rotational-constant diagnostics for the QC reference geometry and the final MORPHEUS semiexperimental geometry. Errors are computed against vibrationally corrected experimental constants for the five-isotopologue data set.

Geometry	RMS/MHz	Max/MHz	RMS/%	Max/%
QC reference	1.7414	3.4072	0.0258	0.0363
SE fit	0.0269	0.0571	0.0005	0.0011

TABLE IX. Parent glycine I rotational constants from QC and MORPHEUS/SE geometries. Corrected experimental and calculated values are in MHz.

Component	Exp. B_e	QC	QC–Exp.	SE	SE–Exp.
A	10420.1910	10422.9024	2.7114	10420.2481	0.0571
B	3907.4785	3907.5931	0.1146	3907.4786	0.0001
C	2934.6809	2935.7175	1.0366	2934.6808	-0.0001

decreases from 1.02×10^4 to 8.02×10^3, and the complete rotational RMS changes only from 0.0269 to 0.0389 MHz. The propagated uncertainties remain chemically meaningful: the largest reported bond-length uncertainty is 0.0017 Å and the largest valence-angle uncertainty is 0.12°. Thus, for a high-quality PCS4 starting geometry, predicates act as a conservative stabilization layer rather than as hidden constraints.

The Gly-I comparison is intentionally stringent: the refinement starts from an already spectroscopically consistent QC structure rather than correcting a poor starting point. The largest heavy-atom bond and valence-angle shifts relative to the final semiexperimental structure are 0.0034 Å and 0.25°, respectively, both within the scale expected for state-of-the-art QC structures and vibrational corrections.[2,11]

Glycine II is the non-symmetric reduced-dimensionality case. The available isotopologues do not determine a stable full structure, so the QC reference is used to generate coupled primitive-coordinate constraints: C–H and N–H differences, two methylene angular combinations, and an average H–N–C–C torsion. These constraints are then projected onto the 24-dimensional C_1 GIC space.

Table X shows that QC-derived targets produce the same rank and essentially the same residual scale as the published Kasalová-type constraints. The final QC-constrained refinement has 19 effective variables, a condition number of 1.05×10^4, and a complete rotational

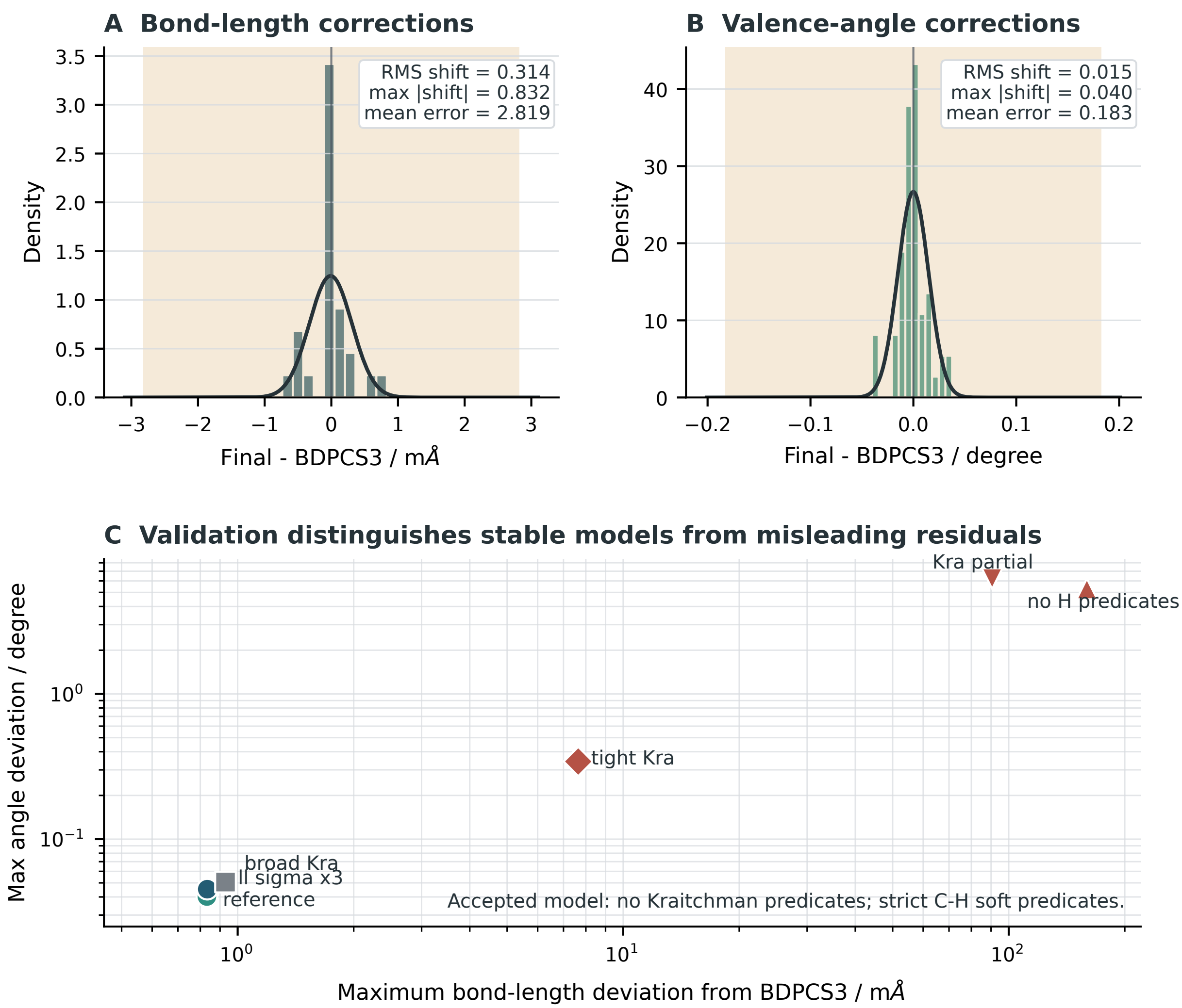


FIG. 7. Camphor validation stress test. Panels A and B show the signed final-minus-BDPCS3 shifts for bond lengths and valence angles in the accepted no-Kraitchman model; the curves are normal distributions fitted to the signed shifts, and the shaded intervals indicate the mean propagated standard error. Panel C compares selected stability variants by the maximum structural deviation from the BDPCS3 reference. The accepted model and predicate-width perturbations remain in the sub-milliangstrom, few-hundredths-of-a-degree region, whereas models that remove hydrogen predicates or impose tight Kraitchman-derived predicates can yield superficially acceptable rotational residuals while distorting the geometry.

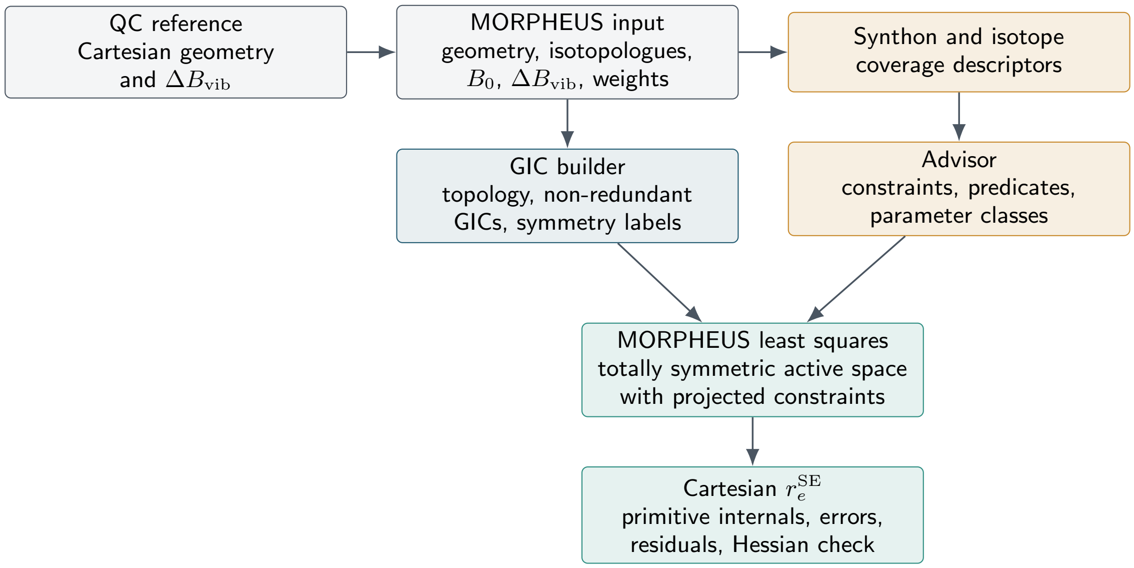


FIG. 8. End-to-end glycine refinement starting from QC Cartesian reference geometries and vibrational corrections. The calculation uses the common job/xyzin container, chemical descriptors, and isotope coverage to generate the active coordinate and constraint model without manually selected fit variables.

TABLE X. Glycine II constraint-source comparison. Both calculations used GIC coordinates, moments of inertia as observables, and the same 10 corrected Gly-IIn rotational data sets. Rotational residuals are reported over the diagnostic A, B, and C constants in MHz. The last column is the mean-square convention used for comparison with the published glycine refinement, not a second RMS value.

Constraint target source	Rank	RMS	Max	$10^3\langle\Delta B^2\rangle$
Kasalová-type values	19	0.2344	0.8100	54.94
QC-derived values	19	0.2491	0.6707	62.06

RMS residual of 0.2491 MHz. The largest uncertainties remain in the amino-hydrogen frame, as expected from the substitution pattern.

Adding the same weighted predicate layer to the QC-constrained Gly-II model preserves the rank at 19, lowers the condition number to 8.55×10^3, and gives a complete rotational RMS of 0.2401 MHz. The heavy-atom bond uncertainties remain below 0.0024 Å and the heavy-atom valence-angle uncertainties below 0.25°, whereas several amino-hydrogen angles and torsions retain larger propagated uncertainties. MORPHEUS therefore reports the predicates as useful stabilization and diagnostics, but does not interpret the poorly substituted hydrogen-frame directions as fully determined structural parameters.

C. Testosterone: Large-Molecule Scale Demonstration

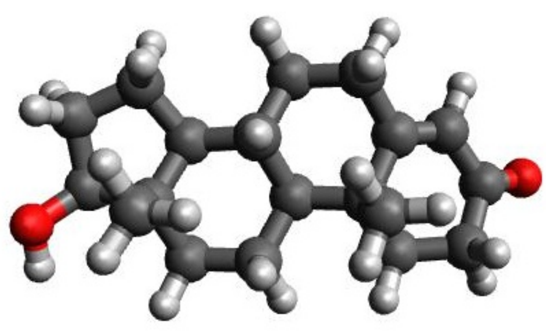
The validation benchmarks above were selected because they represent available experimental gold standards while spanning acyclic, cyclic, fused, bridged, planar, homologous, and reduced-dimensionality cases. The model-generation layer is not limited to these sizes. As a scale check, the framework was also applied to testosterone, using the published experimental ground-state rotational constants and vibrational corrections for the observed T1-H2$^-$-gp conformer.[33] This is a useful large molecule test because this level of treatment is rarely available for systems of this size: an accurate QC equilibrium geometry, accelerated ΔB_{vib} corrections, and algorithmic model generation are combined in one reproducible protocol.

The unrefined QC+ΔB_{vib} residuals with respect to the corrected experimental constants are 4.638, 0.674, and 0.563 MHz for A, B, and C, respectively. After a 141-dimensional non-redundant space is built, the class layer applies a single graph-defined C–C skeletal stretch class. Operationally, non-skeletal stretch combinations and the angular, torsional, and out-of-plane families are held fixed for this parent-only diagnostic, while the retained C–C stretch block is replaced by one shared variable af-

TABLE XI. Effect of weighted predicates on the glycine reduced-dimensionality fits. The predicate runs use the same experimental data and hard functional constraints as the corresponding benchmark runs, plus initial-geometry predicates with $\sigma_R = 0.003$ Å, $\sigma_A = 0.3°$, and $\sigma_D = 0.5°$. RMS values are complete rotational residuals in MHz.

Model	Rank	Cond.	RMS
Gly-I constraints	11	1.02×10^4	0.0269
Gly-I + predicates	11	8.02×10^3	0.0389
Gly-II QC constraints	19	1.05×10^4	0.2491
Gly-II + predicates	19	8.55×10^3	0.2401

ter rank reduction. The resulting refinement improves the residuals to −0.027, 0.633, and 0.292 MHz, reducing the RMS rotational residual from 2.73 to 0.40 MHz. Two additional end-to-end variants test more informative model formulations without changing the input data. Separating C–C and C–O stretch classes gives residuals of 0.002, 0.046, and −0.058 MHz. In this run the model is specified only through primitive C–C and C–O bond classes; MORPHEUS maps the generated GICs onto disjoint shared classes with default coefficient thresholds and an automatic data-compatible class budget, freezing unsupported directions. A predicate model that keeps a compact C–C/C–O subset independent and constrains the corresponding XY QC bond lengths with 0.005 Å uncertainties gives 4.128, 0.704, and 0.555 MHz. Relaxing the same XY predicates to 0.01 Å improves these residuals to 3.114, 0.755, and 0.531 MHz, but still does not approach the two-class C–C/C–O result. Predicate widths should reflect the expected accuracy of the QM reference: for a good equilibrium geometry, 0.005–0.01 Å is already a conservative XY-bond range, whereas 0.05 Å would correspond to an unrealistically loose structural prior. The test therefore shows that one parent isotopologue is too little information for independent predicate-guided C–C/C–O corrections, and that the class model is the appropriate reduced description in this limiting case. With several isotopologues, even when the data set remains incomplete, the glycine tests show that classes and predicates can be combined more effectively. This test is deliberately not presented as an isotope-rich structural validation: it demonstrates that a large-molecule QC geometry, vibrational corrections, non-redundant coordinate generation, class constraints, and weighted predicates can be chained for a 49-atom fused-ring molecule, where sufficiently rich isotopic substitution data are not presently available for a full validation of the fit engine.



Testosterone large-molecule scale demonstration.

Residuals are relative to the corrected experimental equilibrium constants derived from the published B_0 and ΔB_{vib} values.

Model	Variables	ΔA	ΔB	ΔC
QC+ΔB_{vib}	0	4.638	0.674	0.563
C–C class	1	-0.027	0.633	0.292
C–C/C–O classes	2	0.002	0.046	-0.058
XY QC predicates, 0.005 Å	7	4.128	0.704	0.555
XY QC predicates, 0.01 Å	7	3.114	0.755	0.531

VI. CONCLUSIONS

The central contribution of this work is a general framework for proposing and validating constrained refinement models, demonstrated here through semiexperimental structure refinement. Existing refinement methods generally assume that the coordinate model has already been supplied; here topology perception, symmetry assignment, homogeneous rank pruning, constraint projection, least-squares solution, covariance propagation, and final model-stability checks are part of one calculation rather than separate user decisions.

The benchmarks show that the resulting models remain graph-defined, symmetry-consistent, rank-consistent, and non-redundant for acyclic, cyclic, planar, fused, bridged, homologous anhydride, glycine, and large fused-ring cases. The program selects the active symmetry space, removes redundant coordinates, chooses the independent moment components for planar molecules, generates constraints when isotopic coverage is incomplete, and propagates covariance to the reported structural parameters. Where both active coordinate models are applicable, GIC and symmetry-Cartesian refinements give the same effective dimensions and rotational residuals.

Electronic-structure information is used only where the experiment does not fully determine the structure. For Gly-I the QC reference already lies within the practical rotational-accuracy window, so the fit provides a statistically consistent semiexperimental correction and uncertainty estimate. For Gly-II and norcamphor, the advisor converts reference information and isotope coverage into explicit reduced-dimensionality constraints. The reference therefore supplies constraints for unsupported directions, while the spectroscopic data determine all supported directions.

The norbornane-derived series adds the decisive validation step. Norcamphor shows that a bridged low-symmetry topology can be reduced to a stable semiexperimental model when isotope coverage is incomplete. Camphor keeps the same structural motif but adds methyl substitution and a denser set of hydrogen-sensitive local directions. In that harder case, a model with strict BDPCS3-derived C–H predicates and no Kraitchman predicates gives a sub-milliangstrom correction to the reference geometry and remains stable under coordinate-basis changes, robust losses, predicate-width scans, and multistarts. Alternative models that look numerically attractive from the rotational residual alone fail chemically when hydrogen predicates are removed or Kraitchman-derived restraints are made too tight. This is the practical reason for making model generation auditable: the residual is necessary, but it is not a sufficient certificate of a meaningful structure.

This is not simply a convenient input generator. The final outputs remain ordinary chemical objects: Cartesian structures, bond lengths, valence angles, dihedrals, ring descriptors, propagated uncertainties, residuals, and diagnostics. What changes is that the path from topology and data to those objects is explicit, reproducible, rank-audited, and challenged by independent stability checks, even for molecular graphs in which manual coordinate formulation is often the limiting step.

The testosterone scale check extends this logic to the current upper end of practical rotational-spectroscopic modeling. It does not claim a complete semiexperimental structure from a parent-only data set; instead, it shows that an accurate large-molecule QC geometry, accelerated ΔB_{vib} corrections, algorithmic non-redundant GIC generation, and class-constrained refinement can be chained for a 49-atom fused-ring molecule in a reproducible calculation. This closes the loop from accurate large-molecule QC structures to algorithmically generated spectroscopic refinement models.

The essential shift is from expert-dependent model formulation to reproducible algorithmic model generation integrated with the refinement itself. The algorithm records and enforces chemical judgement in an auditable form rather than replacing it with hidden manual choices. Although motivated by semiexperimental equilibrium structures, the underlying graph-driven strategy is applicable to any constrained refinement problem requiring simultaneous treatment of topology, symmetry, rank deficiency, and external structural information.

SUPPORTING INFORMATION AVAILABLE

Atom-numbered structures for the benchmark and companion systems; complete rotational-constant comparisons for glycine I, glycine II, cyclopentadiene, nitrobenzene, and azulene; Kraitchman, reference-fit, and planar-component diagnostics; additional diagnostic tables for glycolaldehyde, norcamphor, and camphor; final Cartesian coordinates, propagated topological parameters, residuals, and archived `MORPHEUS` output files.

CODE AVAILABILITY

`MORPHEUS` is being integrated into the broader `MATRIX` framework. The version used for the calculations reported here will be made available shortly as part of the `MATRIX` distribution, together with the benchmark inputs and audit manifests needed to reproduce the tables and figures.

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